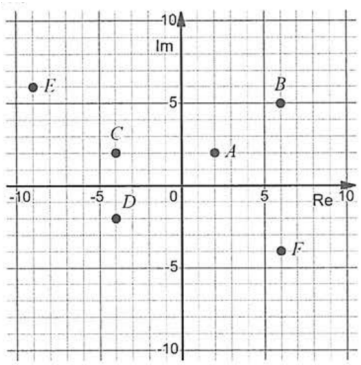
Complex Numbers: Argand Diagrams

Questions 1–27

Question 1. Write down the complex numbers represented by each of the points in the Argand diagram below:



Question 2. Represent each of the following on an Argand diagram. Use a separate diagram for each part:

a) $z = 2 - 3i$ and $w = -5 + 2i$
b) $z = 3i$ and $w = 1$
c) $z = -4 + i$ and $-z$
d) $z = -1 - i$ and z^*
e) $z = 1 - 3i$, $2z$ and iz

Question 3. Represent the solution of each of the following quadratic equations on an Argand Diagram:

a) $z^2 + 2z + 4 = 0$
b) $z^2 - 3z + 3 = 0$
c) $z^2 - z + 5 = 0$
d) $z^2 + 14z + 80 = 0$

Question 4. Given that $z = 2 + i$, for each of the following add the point to a single argand diagram and describe the transformation that takes z to the given point:

a) z^*
b) $-z$
c) iz
d) $-iz$

Question 5. Given that $z = 3 - i$, for each of the following add the point to a single argand diagram and describe the transformation that takes z to the given point:

a) z^*
b) $-z$
c) iz
d) $-iz$

Question 6. Given that $z = -5 - 2i$, for each of the following add the point to a single argand diagram and describe the transformation that takes z to the given point:

a) z^*
b) $-z$
c) iz
d) $-iz$

Question 7. Given that $z = 1 + 3i$ and $w = 2 + 2i$, on a single diagram plot each of the following:

a) z
b) w
c) $z + w$
d) $z - w$

Question 8. Given that $z = -4 + 5i$ and $w = 5 - 2i$, on a single diagram plot each of the following:

a) z
b) w
c) $z + w$
d) $z - w$

Question 9. Given that $z = 3 + i$

a) Show that $z^2 = 8 + 6i$

b) Plot z and z^2 on the same Argand diagram.

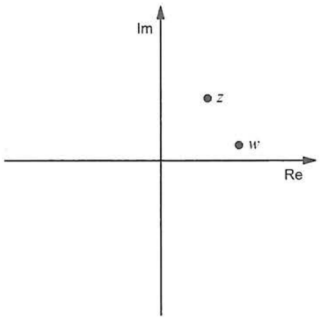
Question 10. Given that $z = -1 + i$

a) Find z^2, z^3, z^4 and z^5

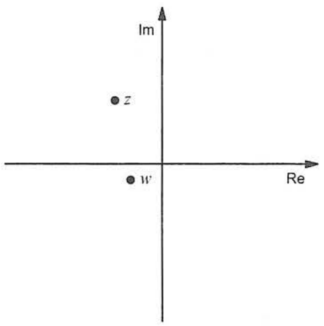
b) Plot z, z^2, z^3, z^4 and z^5 on the same Argand diagram.

Question 11. For each of the following diagrams, copy the diagram and add the points z^*, $-w$ and $z+w$

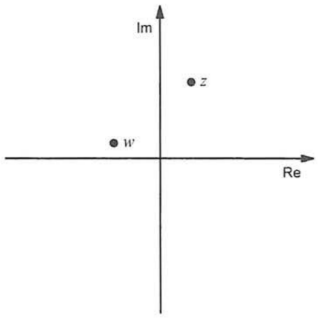
a)



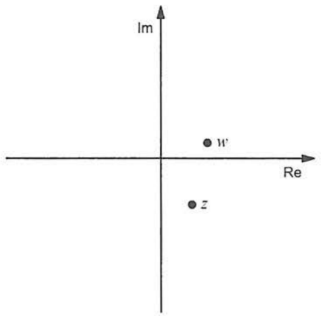
b)



c)



d)



Question 12. Prerequisite knowledge for Q12: Complex Conjugates

Given that $z_1 = 4 - 2i$ and $z_2 = \frac{1+i}{z_1}$

a) Find z_2 in the form $a + bi$

b) Show on an Argand diagram two points representing z_1 and z_2

Question 13. Given that $z_1 = 2 + i$, $z_2 = iz_1$, $z_3 = iz_2$, $z_4 = iz_3$ and $z_5 = iz_4$

a) plot z_1, z_2, z_3, z_4 and z_5 on a single Argand diagram.

b) Describe the mapping that transforms z_1 to z_2

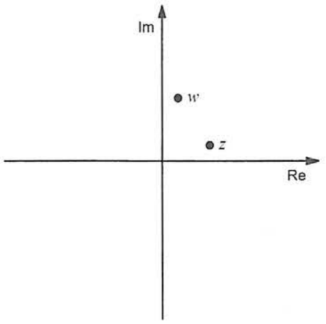
Question 14. Given that $z_1 = -5 + 2i$, $z_2 = -iz_1$, $z_3 = -iz_2$, $z_4 = -iz_3$ and $z_5 = -iz_4$

a) plot z_1, z_2, z_3, z_4 and z_5 on a single Argand diagram.

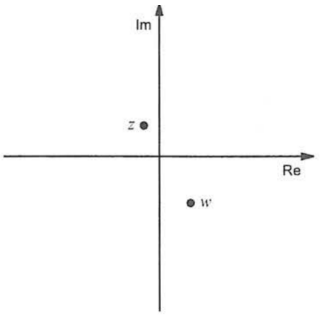
b) Describe the mapping that transforms z_1 to z_2

Question 15. For each of the following diagrams, copy the diagram and add the points z^*, z^2 and $z - w$

a)



b)



Question 16. In an Argand diagram the points representing the numbers $3 + 3i$ and $3 - i$ are two adjacent vertices of a square.

a) Find the area of the square.

b) Find all the possible pairs of numbers represented by the two other vertices.

Question 17. Distances on an Argand Diagrams

The distance between the origin and the complex number $a + bi$ is r where a and b are real constants.

a) Write r in terms of a and b

b) Show that $r^2 = |zz^*|$

Question 18. Two complex numbers are given by $z = a + bi$ and $w = c + di$ where a, b, c and d are real constants. The points z and w are a distance of p and q from the origin respectively.

a) Find zw

b) By considering the lengths in terms of a, b, c and d, prove that $|zw| = pq$

Question 19. Given that $z = 2 - 3i$

a) find iz

b) Find $(1 + i)z$

c) Prove that the shape formed by the origin, z, iz and $(1 + i)z$ is a square.

d) State the area of the square.

Question 20. It is given that $z = a + bi$ for constants a and b

a) Find iz

b) Find $(1 + i)z$

c) Prove that the shape formed by the origin, z, iz and $(1 + i)z$ is a square.

d) State the area of the square.

Question 21. The points on an Argand diagram P and Q are given by $z = a + bi$ and iz respectively, where a, b, c and d are real constants.

a) Find the length of OP and OQ

b) Prove that OP is perpendicular to OQ

Question 22. The points on an Argand diagram A, B and C are given by

$$z = 2 + i, \ z^3 \text{ and } 2z + z^3$$

respectively.

a) Plot the points A, B and C on an Argand diagram.

b) Find the length of the line segment OA

c) Find the length of the line segment BC

d) Describe the relationship between the length of the line segment OA and that of the line segment BC

Question 23. The points on an Argand diagram A, B and C are given by

$$z = 1 + 2i,\ z^* \text{ and } 3z + z^*$$

respectively.

a) Plot the points A, B and C on an Argand diagram.
b) Find the length of the line segment AC
c) Find the length of the line segment BC
d) Find the area of triangle ABC

Question 24. Given that $z = 1 + i$

a) find z^2, z^3, z^4 and z^5
b) Plot z, z^2, z^3, z^4 and z^5 on the same Argand diagram.
c) Find the distance between each of z^2, z^3, z^4 and z^5 and the origin
d) Hence, state the distance between z^n and the origin where n is an integer.

Question 25. Prerequisite knowledge for Q25: Complex Conjugates
It is given that

$$z = -20 + 21i \text{ and } w = 5 + 2i$$

a) Show that $\frac{z}{w} = -2 + 5i$
b) Plot w and $\frac{z}{w}$ on an Argand Diagram.
c) Prove that the origin and the points w and $\frac{z}{w}$ form a right-angled triangle.
d) Find the area of the triangle identified in part c).

Question 26. The points on an Argand diagram A and B are given by

$$z = 2 + 5i \text{ and } w = -2 + i$$

respectively.
Show that the area of triangle OAB is 6 units^2.

Question 27. The points on an Argand diagram A and B are given by $z = a + bi$ and $w = c + di$ respectively, where a, b, c and d are real constants.
Show that the area of triangle OAB is given by

$$\frac{1}{2} \times |\operatorname{Im}(zw^*)|$$